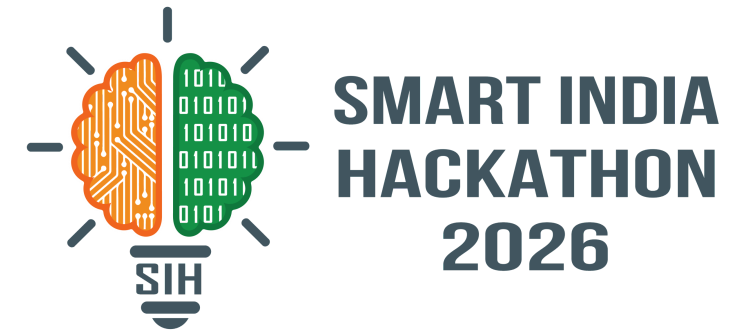


SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH26102**
- **Problem Statement Title - Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Scheme implementation regd.**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID- 135230**
- **Team Name- WEBSHASTRA**



PROBLEM STATEMENT

- MPLADS funds are used by MPs for development projects in their constituencies.
- The government needs to monitor these projects and check how the funds are being used.
- The challenge is to detect irregularities, fraud, duplicate works, cost issues and other risks.

PROPOSED SOLUTION

- AI Evidence Verification Layer over eSAKSHI
- Scans 45,000+ works & payments across India
- Calibrated Composite Risk Score (0–100)
- Structured Explainable Evidence Cards
- 1-Click CAG/DM Investigation Brief

OUR AI WORKFLOW

E- SAKSHI / MPLADS DATA

CROSS-DATA LINKING

• WORK ID • VENDOR
• AGENCY • TIMELINE

AI ANALYSIS

• ML • NLP
• CV Graph • Rules

RISK FUSION

RISK SCORE (0-100)

EVIDENCE +
EXPLANATION

STRUCTURED CARDS

 INVESTIGATION BRIEF



REAL WORLD IMPACT

- Curbs public fund leakage & ghost assets
- 80% reduction in manual audit workload
- Unblocks stalled civic infrastructure
- Guarantees mandatory 15% SC & 7.5% ST quota compliance



INNOVATION

- 21-Module Multimodal AI (Rules + ML + NLP + CV + Graph)
- All mini LM for duplicate & split works
- Bipartite Graph HHI for contractor cartel detection
- Statutory Grounding in 2023 Guidelines

TECH STACK

Data Ingestion

12 eSAKSHI CSVs
Vendor Ledgers
Guidelines PDF

Multimodal AI core

All mini LM L6 V2
MAD Cost Outliers
CV Geotagged
Image Match

Graphs and predictions

Bipartite
Network Graph
XGBoost
Failure Predictor
HHI Cartel Index

XAI and Copilot

SHAP Attribution
Evidence Cards
RAG PDF Citations

Foundations

 **PYTHON**
Core development

 **PANDAS**
Data Analysis

 **NumPY**
Numerical Computing

 **SciPY**
Scientific Computing

ML & ANALYTICS

• RapidFuzz • NetworkX • Perceptual Image Hashing (dHash) • Multi-Vector Risk Fusion

BACKEND & API

Node.js (Express.js Gateway & RBAC) • Python (FastAPI AI Engine) • Supabase PostgreSQL • Supabase Auth & Storage CDN

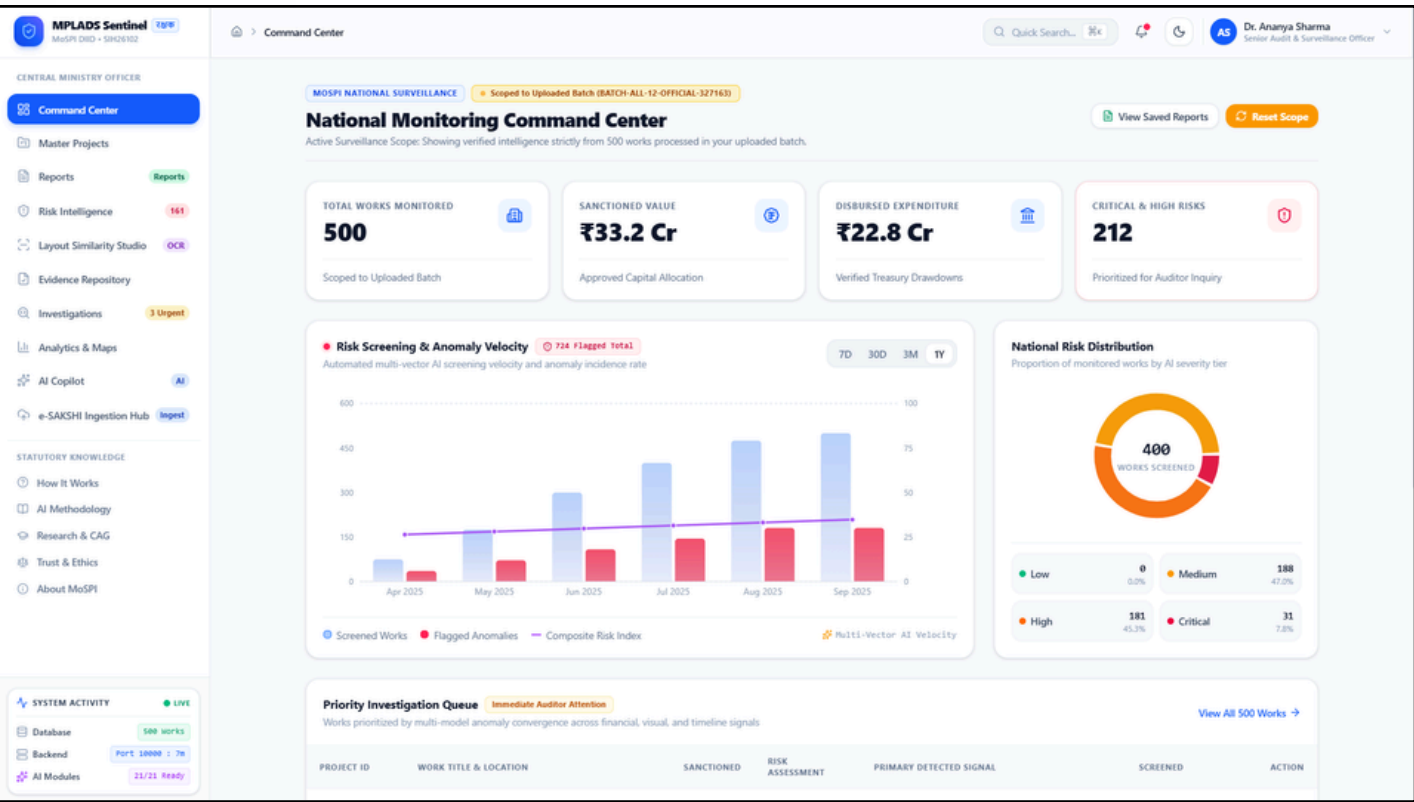
FRONTEND & GIS

Next.js 16 • React 19 • Tailwind CSS v4 • Recharts • Haversine GIS Spatial Buffering & Proximity Clustering

XAI & COPILOT

Google Gemini 2.0 Flash • Grounded RAG (MPLADS Guidelines 2023 & GFR 2017) • Multi-Source Evidence Lineage

WEBSITE MOCKUP



MULTI-DIMENSIONAL FEASIBILITY

1

FINANCIAL

100% open-source stack, ₹0 licensing fees, massive ROI by preventing multi-crore fund leakages.

2

OPERATIONAL

Non-intrusive surveillance layer over eSAKSHI without altering officer workflows.

3

TECHNICAL & DATA

Uses proven lightweight ML models on commodity CPUs; ingests real 45,000+ public records.

KEY CHALLENGES & MITIGATION

ABSENCE OF
FRAUD LABELS



Solved using unsupervised anomaly detection + statistical benchmarks rather than unverified classifiers.

FALSE
POSITIVES

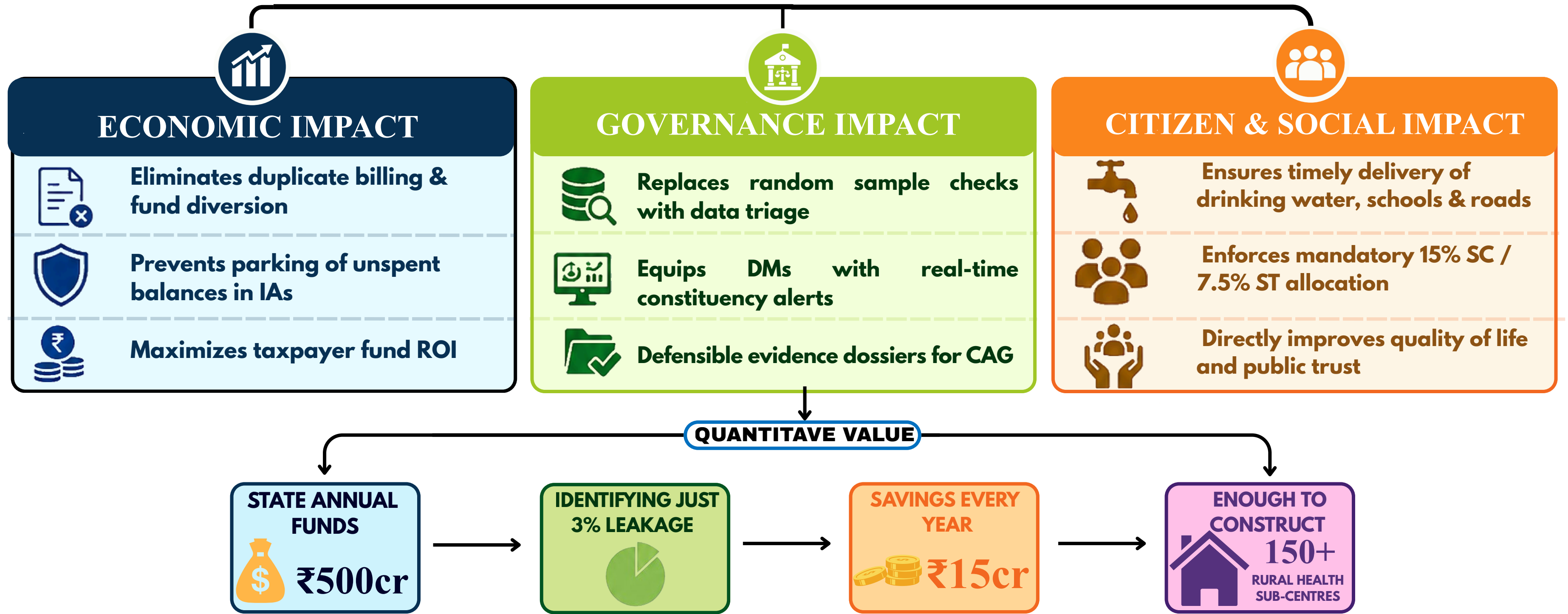


Multi-signal calibration ensures works are only flagged critical if ≥ 2 independent signals trigger.

POLITICAL
SENSITIVITY



Framed as an objective "Investigation Priority Index" with evidence cards rather than accusations.



1. **Comptroller and Auditor General of India (CAG)**: Compliance Audit on Implementation of Members of Parliament Local Area Development Scheme, Report No. 4 .
2. **Ministry of Statistics and Programme Implementation (MoSPI)**: MPLADS Scheme Guidelines (Revised February 2023) — Operationalization of Central Nodal Agency (CNA).
3. **L. Potin, R. Figueiredo, V. Labatut, C. Langeron**: Pattern Mining for Anomaly Detection in Graphs: Application to Fraud in Public Procurement, arXiv preprint arXiv:2306.10857, 2023.
4. **J. Bai, T. Qiu**: Automatic Procurement Fraud Detection with Machine Learning, arXiv preprint arXiv:2304.10105, 2023.
5. **S. Lundberg & S. Lee**: A Unified Approach to Interpreting Model Predictions (SHAP), NeurIPS.